Pen Test Pocket

Draft 2 Report

Dayton Dekam - CrowdStrike

Mentor:
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PTP Adelson

Solo Project
Zachary Adelson

CptS 432 Cybersecurity Capstone Project

Fall 2025

Note: Recall that this writing assignment should follow these guidelines:

Length = minimum of 5 pages text + appendixes as needed - though, this should be *MUCH* longer than 5 pages if you leverage all of your prior documents in full. There's no maximum page limit for this document, but be judicious in your screenshots that just fill space.

Sections that do not count to content for page limit:

- Cover page
- table of contents
- pictures
- tables
- images
- diagrams

As shown in the Syllabus, this assignment carries a weight of 20% for the final course score.

Posted as a single self-contained file (no links to outside resources that should be directly in this document itself. A link to your git repo == good. A link to something that says "see here for our detailed description"… not as cool.)

Posted as a PDF file.

Typed single-spaced.

Typed with black text.

Typed with #11 font size.

Typed using Arial font.

Typed with one inch margins on sides, top and bottom.

<u>Please erase this page in your final document.</u>

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I. Introduction

I.1. Project Introduction

As a student and upcoming professional in Cybersecurity, it is essential to always and consistently be upskilling, by learning new tools and skills. One of the biggest hurdles I have found when it comes to learning and using tools is remembering all the possible input flags for command line-based interaction. The **PenTestPocket** (**PTP**) solves the struggles that are associated with learning tools, and then not only simplifies the process but improves on the experience with the addition of new features.

To accomplish these goals, I created GUI based toolkit rather than the started CLI based ones, which allows users to fully interact with and use common pen testing tools such as Nmap, NetCat, etc. While there are many tools available as web applications that allow for tools similar to **PTP**, such as nmap.online or RapidTables Conversion tools, that can each individually fulfill the goal of the **PTP** in making these tools and these tasks easy to understand and view without requiring working with a CLI. **PTP** congregates all these tools into a single tool, so no jumping around between websites and tools, everything you will need will always be right in front of you.

Some of the additional features that **PTP** has to offer are logging of all actions and test results for later review, pinning of tools to the homepage of the application for ease of access and use, and a built-in written guide for all the tools in question. These features will enable larger scale tests to be automatically documented, and all that would be required of the end user is reformatting and copying from the user log files which are encrypted within the file but will be able to viewed in clear text within **PTP** so only authorized users can access the log files for confidentiality purposes. We will describe how **PTP** maintains the CIA triad later in this report.

1. My primary client is Dayton Dekam from CrowdStrike, where he agreed to represent as the professional sponsor of the project, but this project is directly affiliated with CrowdStrike. My mentor and professor Ananth Jillepalli serves as the main point of communication for the project. The final project will highlight the skills and capabilities offered by **PTP**

2. Dayton Dekam, CrowdStrike, daytondekam@crowdstrike.com

3. According to CrowdStrike's official website "CrowdStrike is a global cybersecurity leader with an advanced cloud-native platform for **protecting endpoints, cloud workloads, identities and data"**. [Unsure of Dayton's role currently]

I.2. Project Developer Bio

Zachary Adelson is a Senior at Washington State University and will graduate with his Bachelor of Science degree in Cybersecurity, a minor in mathematics, and the CySER CAE-CO Fundamentals Certificate in May of 2026. Zachary has always had an interest in technology and aims to make technology the core focus of his professional career. Prior to PTP, Zachary has participated in multiple CTF and Hackathon competitions, has held two internships at two very distinct companies where he worked as an IT technician and a Technology Risk Assurance auditor for each respective company. As this is a solo project, under the guidance of Ananth Jillepalli and the sponsor backing of Dayton Dekam, Zachary is responsible for all aspects of this project, from GUI design, creation, and code development.

II. Project Requirements Specification

From your revised “Project Requirements Specification” document include the following:

II.1. Project Stakeholders

The stakeholders are CrowdStrike, my professor, and myself, but are not restricted to them.

Potential stakeholders can include any cybersecurity professional. As this tool is geared towards enabling an easy and more efficient testing workflow, **PTP** is the perfect project for anyone looking to break into cybersecurity and streamline their process flow.

All stakeholders involved would benefit from having a clear and concise way of performing penetration tests and that all user logging can be used for future personal documentation, which is a key part of any sort of reporting.

II.2. Use Cases

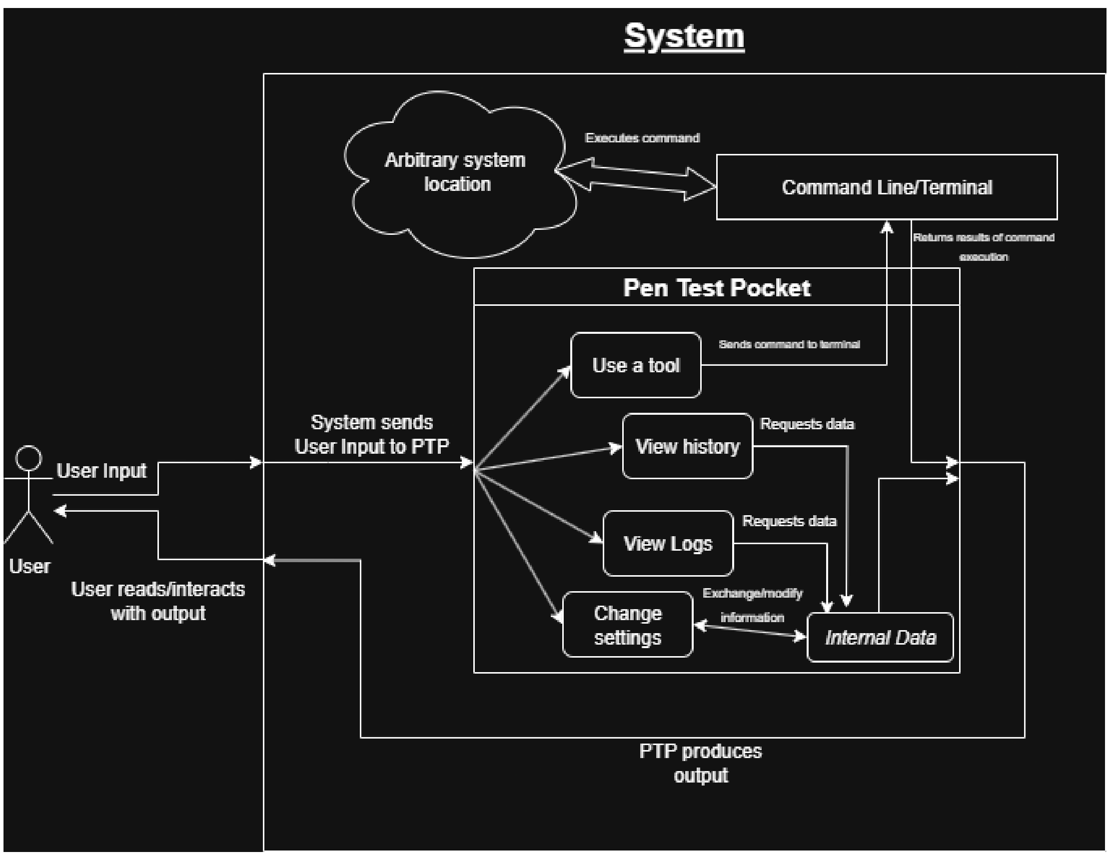


Figure 1: PTP UML Diagram

User Input

Pre-condition	- PTP is open and running
Post-condition	- User input processed - PTP can now act on that input - Events logged
Intended Path	1. Open PTP 2. Select any user interface 3. Interact with said interface to do an action
Alternative Path	
Related Requirements	- Use a tool - View history - View Logs - Change settings

Use a tool

Pre-condition	- User has defined what tool is being called - User has defined all necessary parameters and flag(s)
Post-condition	- Results of tool use received and returned - Tool results logged
Intended Path	1. Selection of tool 2. Passed in parameters and flag(s)
Alternative Path	I. Integrated terminal II. Use tool as if it were CLI based
Related Requirements	- Systems terminal - Systems related resources

View History

Pre-condition	- A tool(s) has been executed
Post-condition	- Specific tool history information displayed
Intended Path	1. Selection of tool within history box
Alternative Path	
Related Requirements	- Internal PTP data

View Logs

Pre-condition	
Post-condition	- All logs are displayed to the user
Intended Path	1. Log button selected
Alternative Path	
Related Requirements	- Internal PTP data

Change settings

Pre-condition	- Settings configurations exist
Post-condition	- Any changes are applied if save is selected
Intended Path	1. Go to settings button
Alternative Path	
Related Requirements	- Internal PTP data

II.3. Functional Requirements

1. Graphical User Interface

Description	The PTP application is designed around having a Graphical User Interface (GUI) for the users to interact with. To be able to achieve this goal I found a python3 based GUI stack, of which I am using PyQt5
Dependencies	Python3, PyQt5, related python3 libraries, all integrated tools (Nmap, hashing tools, etc.)
Categorization	***Code***; as defined on GitHub roadmap

2. Tool Integration

Description	As the pen testing tools are used within the application itself and not in external location such as CLI, it is crucial to the success of the application that the tools be seamlessly incorporated into the PTP application. Tools should be designed to be implemented in a way that allows for ease of modification, removal, or additions
Dependencies	Nmap, netcat, smbclient, snmpwalk, onesixtone, gobuster, curl, whatweb, Metasploit, searchsploit, hashcat, john the ripper, encoding/decoding (binary, hexadecimal, decimal, base64, ascii, utf), SHA-1, SHA-256, MD5, CRC-32, RIPEMD-160, B-crypt
Categorization	***Code***; as defined on GitHub roadmap

3. Terminal Integration

Description	PTP intends to have an integrated terminal so that any terminal-based commands can be run within PTP without the need of opening an independent terminal to run one or two simple commands. The integrated terminal will communicate with the systems terminal to get real and accurate execute results
Dependencies	BASH, PowerShell
Categorization	***Code***; as defined on GitHub roadmap

II.4. Non-Functional Requirements

Succinct GUI:

- The GUI should be simple to understand. There should be little to no learning curve required for utilizing PTP, as that would detract from the overall goal of PTP, shifting learning the tools to learning how to use the tool to simplify the tools.

Ease of Operation:

- Accessing and using any part of PTP should be able to be achieved without much effort. All tool operations should be streamlined such that you'd only need to read the text presented about the tool in order to figure out how to use it.

All Encompassing Needs Met:

- PTP will have every aspect of execution of workflow integrated within the tool itself, including but not limited to; Tool help, tool execution, secure note taking, and secure log files.

Maintaining CIA:

- PTP will maintain all three sectors of CIA in the following ways
    - *Confidentiality* – Using encryption techniques to prevent our user log or note data isn't stored and plain text and cannot be exfiltrated easily by potential threat actors.
    - *Integrity* – By making use of hashing, we can ensure that data within PTP isn't be changed without intention or permission to do so.
    - *Availability* – PTP will ensure that the application is always available when needed and all elements are accessible, and by maintain our confidentiality and integrity we can be sure that our program will always be accessible.

III. Software Design - From Solution Approach

III.1. Architecture Design

PTP was designed with three aspects in mind, security, simplicity, and expandability

III.1.1. Overview

PTP has security as its forefront as the app is designed to test security, it will make sure its threat resilient with the CIA scheme. It is simple in its end user design such that users will be able to use the tool without any additional hassles other than reading the text on the screen in front of them. Lastly, PTP is built to be easily able to add additional tools and alter the code for user cases

III.1.2. Subsystem Decomposition

I broke PTP into a few separate parts; Main page, tool page, settings tab, change log, help window, terminal screen, and user log.

- Main Page
  The main page includes all the pinned tools, connections to all other tabs, and history of tool executions
- Tool Page
  The tool page will pop up whenever a tool is selected, and all its associated attributes will be available for customization based on user needs
- Settings Tab
  The settings tab deals with everything regarding customization of the application including theme, sound settings, terminal customization
- Change Log
  The change log will keep track of and display all changes and additions made to PTP by me throughout the course of its development
- Help Window
  The help window will hold all information regarding use of PTP including navigation of the GUI, individual tool specifications, and how the app works
- Terminal Screen
  The terminal screen is an integrated section of PTP that acts like a real terminal but in the background only acts as a middleman between the user and the system terminal
- User Log
  The user log is where all user events are tracked and stored for later viewing, these can range anywhere from user launching the application, to exactly what tool was run and when

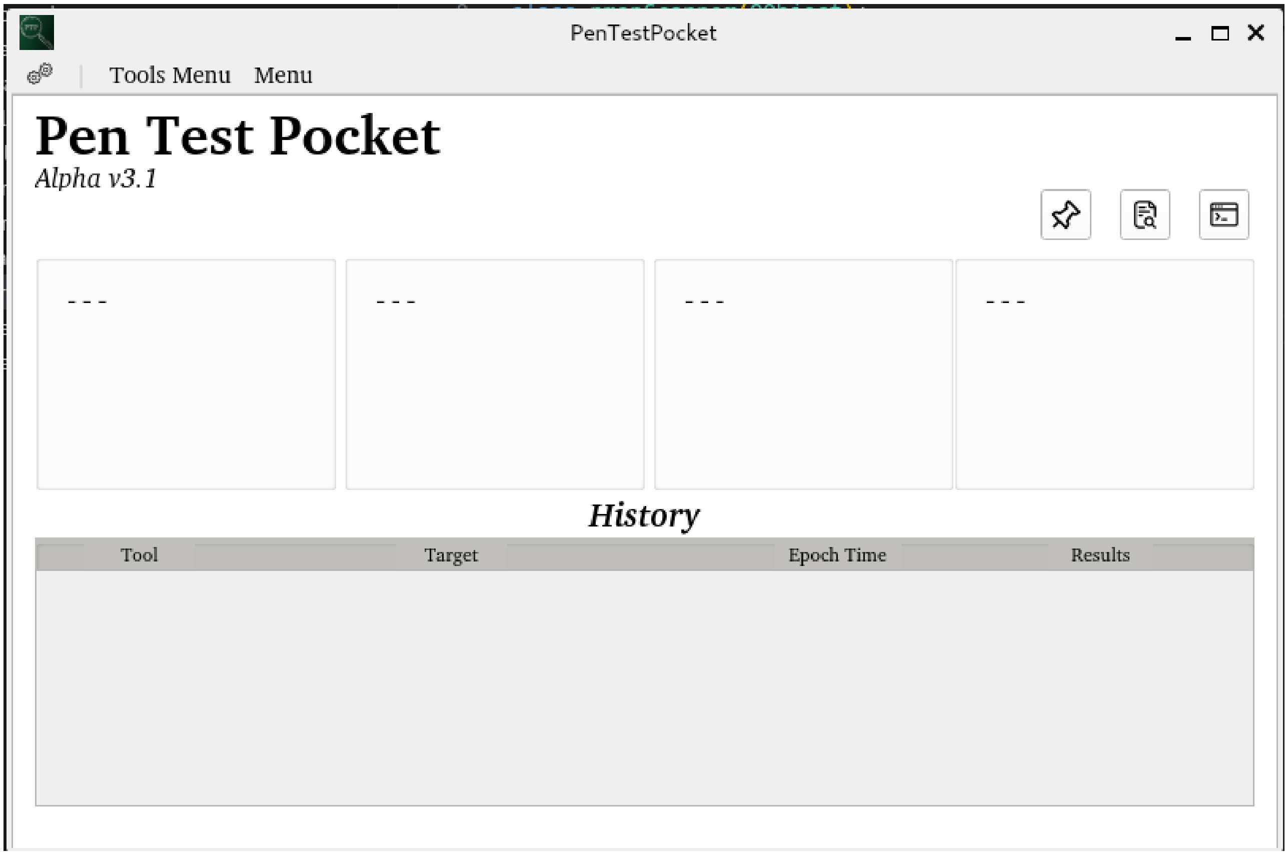
III.2. Data design

Within PTP I use lists, dictionaries, and tuples to store all temporary and persistent information, as well as txt and JSON files for all additional storage that needs to persist between sessions.

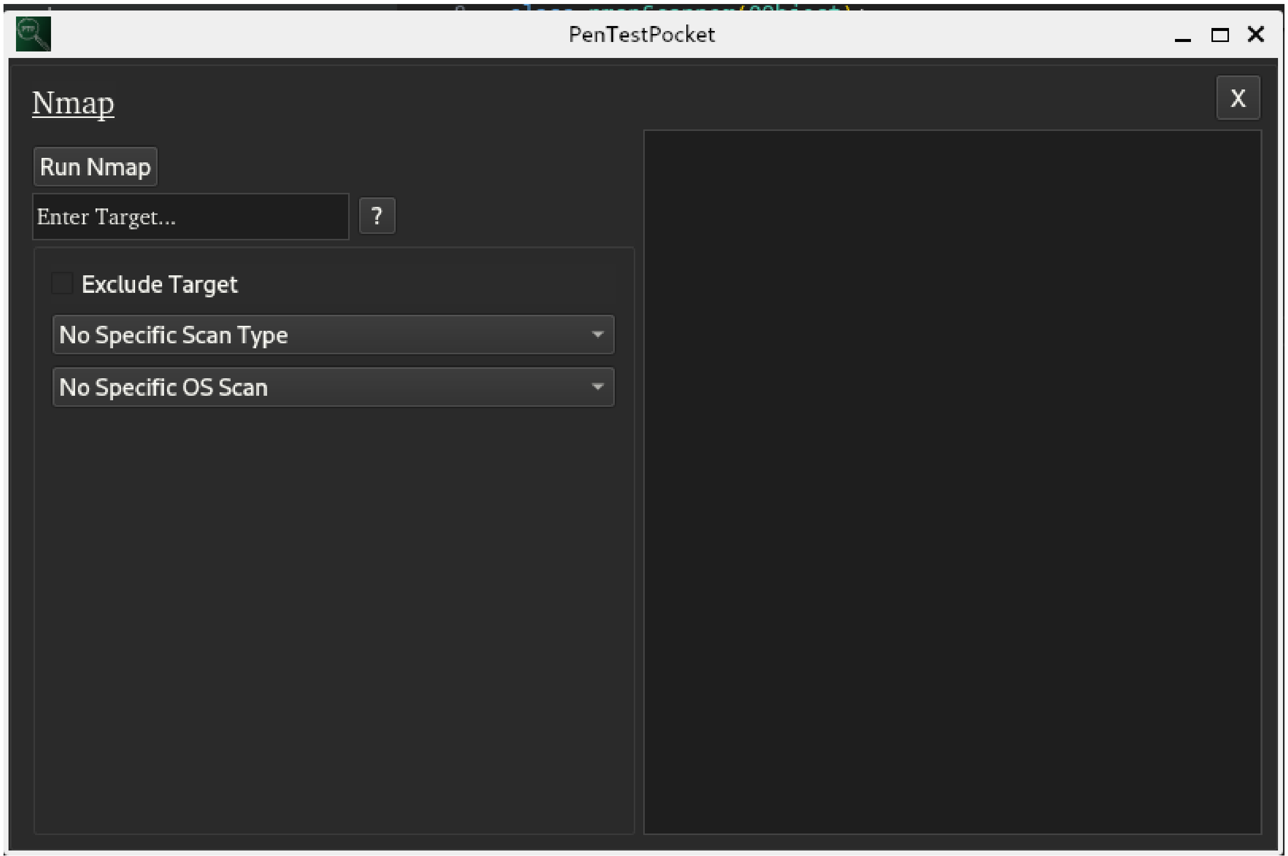
III.3. User Interface Design

The GUI has multiple sections as mentioned above: Main page, tool page, settings tab, change log, help window, terminal screen, and user log. All tools are currently in progress, but each will be documented at its current stage below:

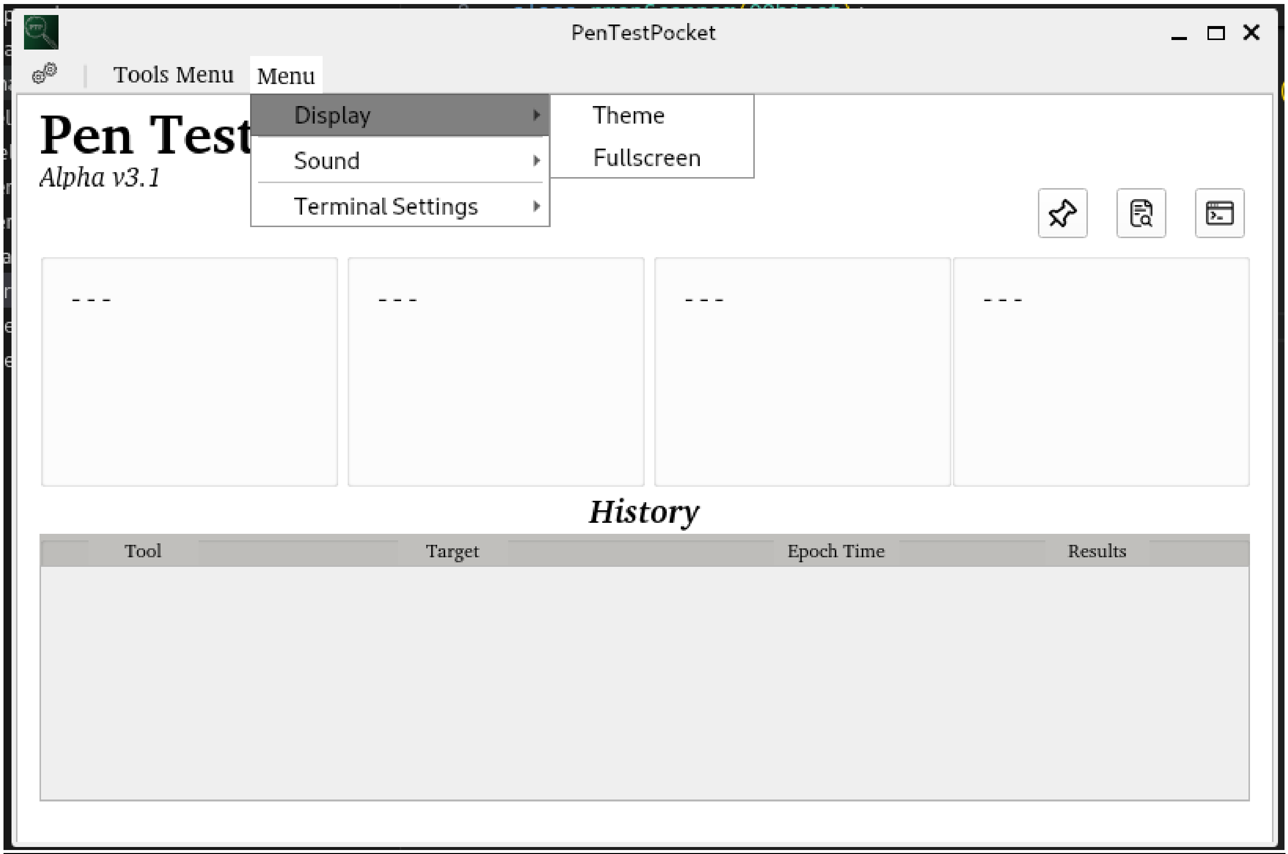
Main Page



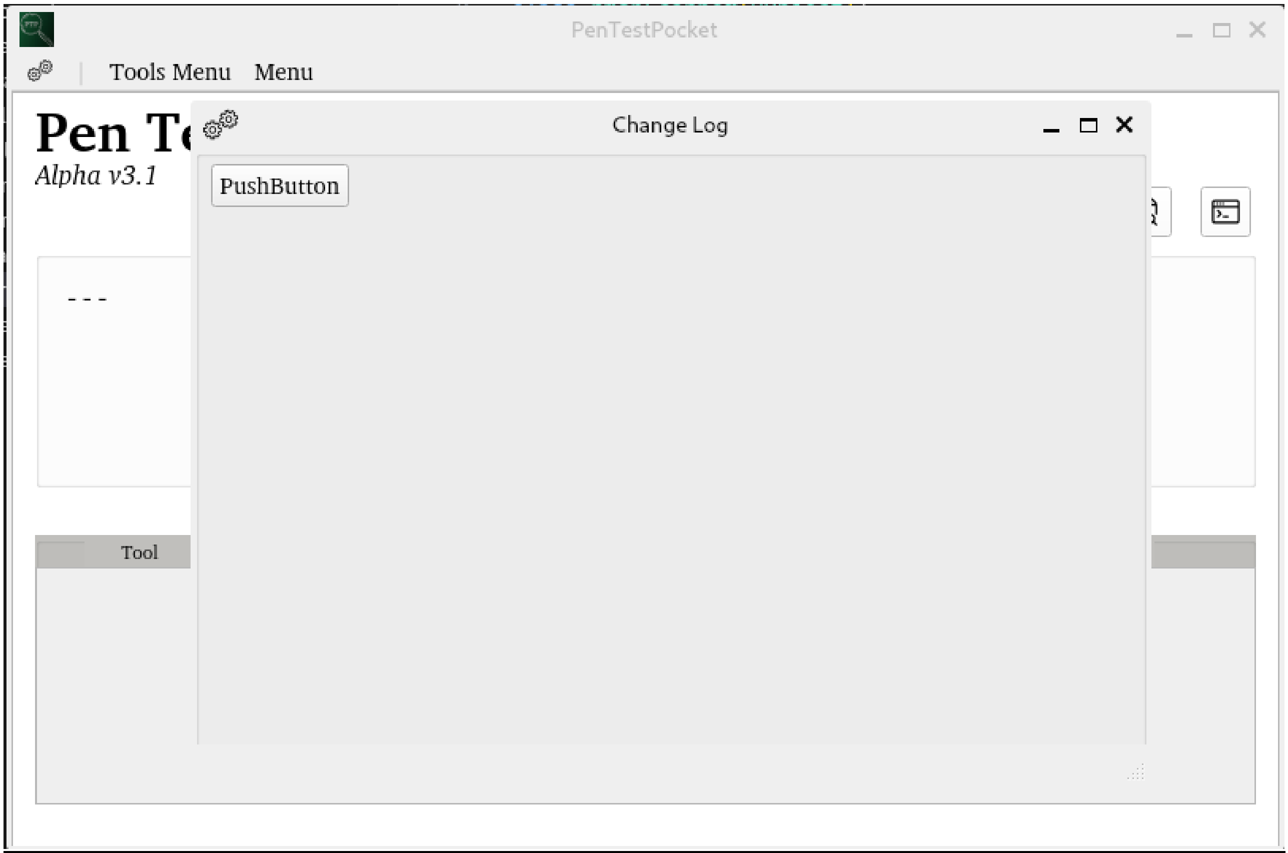
Tool Page


PenTestPocket
Nmap
X
Run Nmap
Enter Target...
?
Exclude Target
No Specific Scan Type
No Specific OS Scan

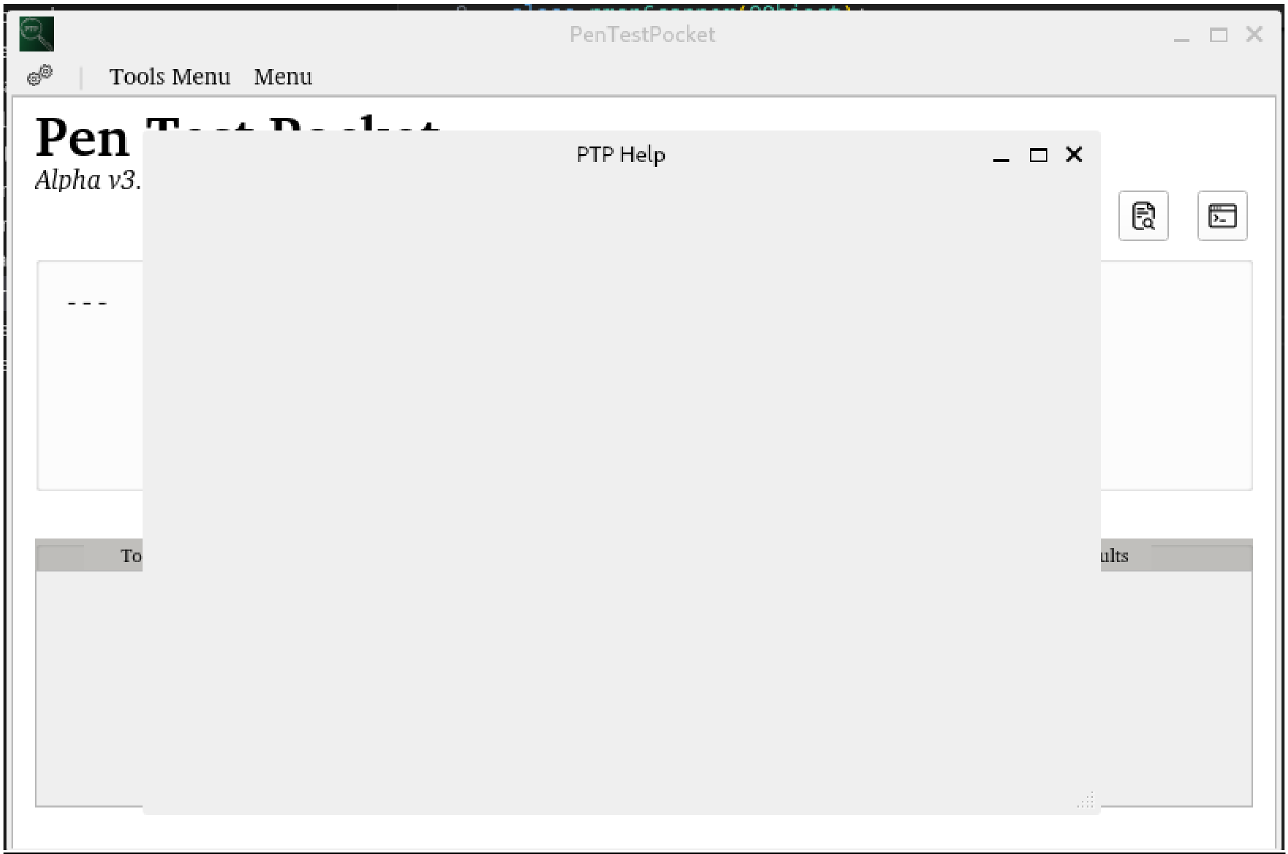
Settings Tab



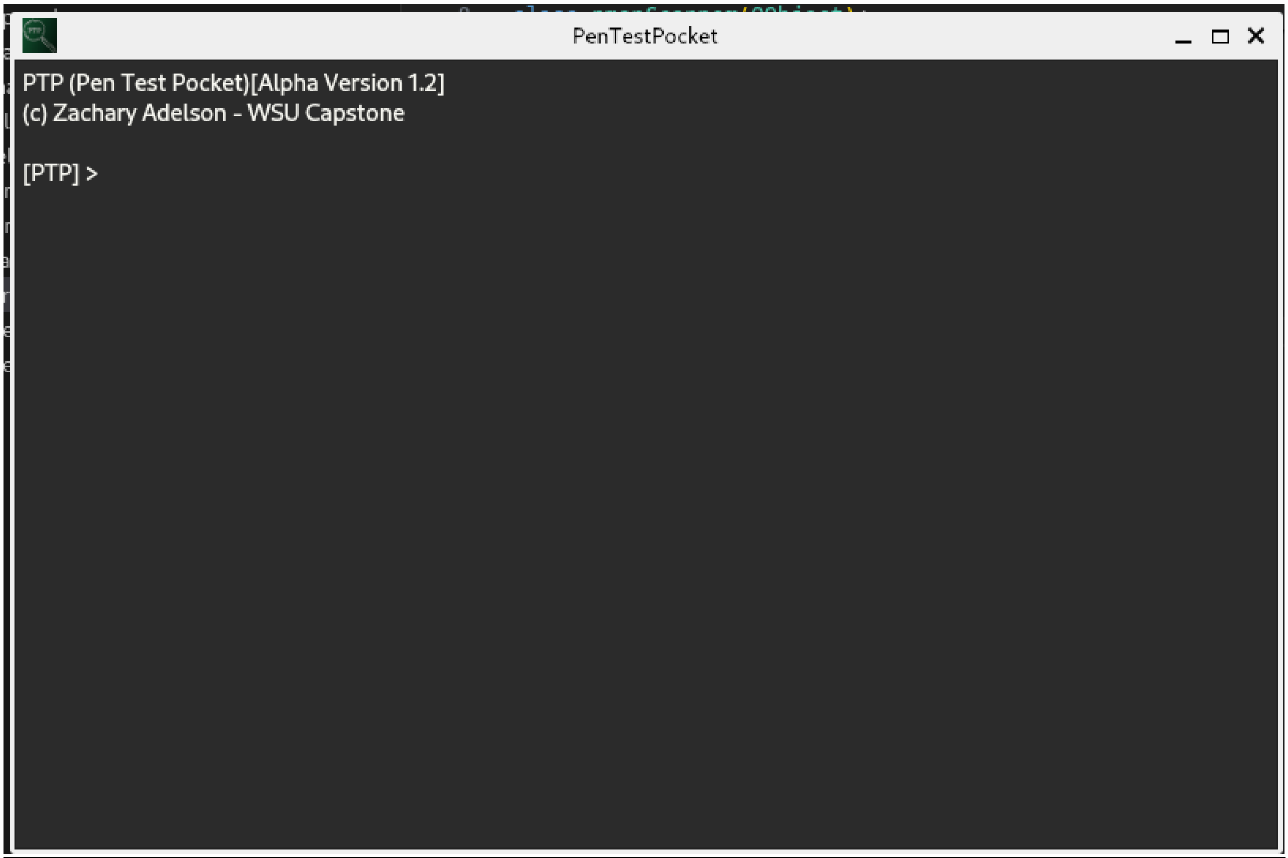
Change Log


PenTestPocket
Tools Menu
Menu
Pen T
Alpha v3.1
Change Log
PushButton
- - -
Tool

Help Window



Terminal Screen



User Log

IV. Test Case Specifications and Results

IV.1. Testing Inputs

<u>Test 1 – Nmap comparisons</u>

a. Test Input – *nmap 127.0.0.1*

b. Test Input – *nmap 8.8.8.8*

c. Test Input – *nmap -exclude 8.8.8.8*

IV.2. Environment Requirements

To support testing, PyQt5 alongside Python3 must be installed, and as the project progresses all additional installed tools will also need to be present, all requirement installation commands are listed on GitHub readme page.

<u>Additional tools</u>

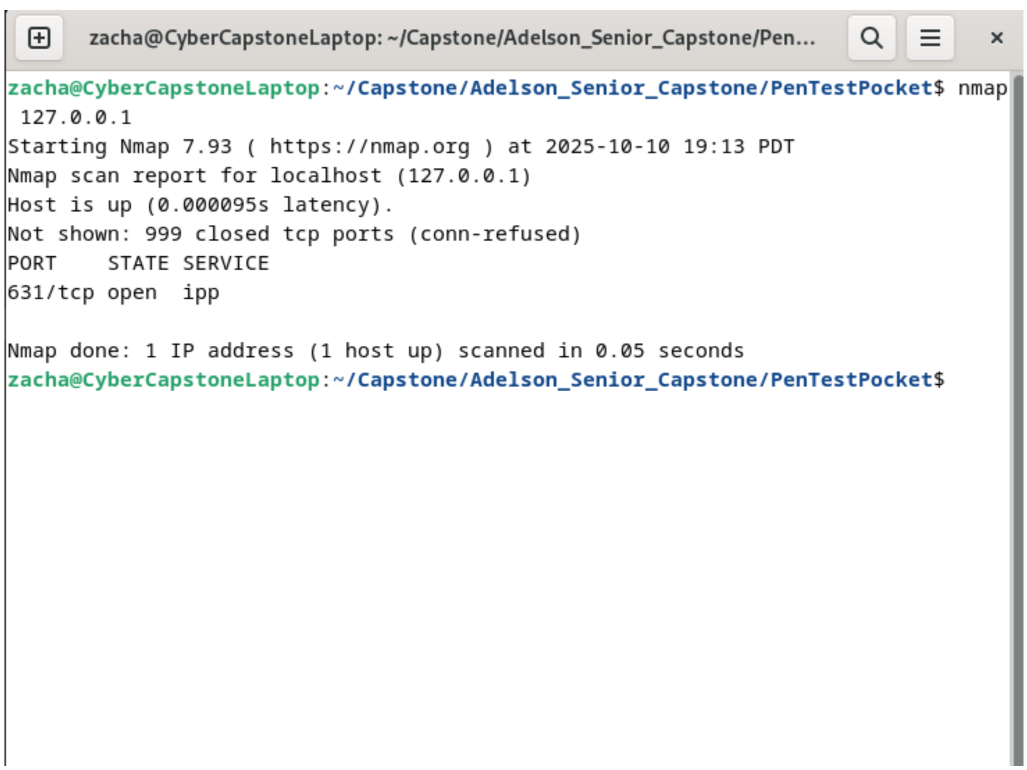
- Nmap

IV.3. Test Results

<u>Test 1 Results</u>

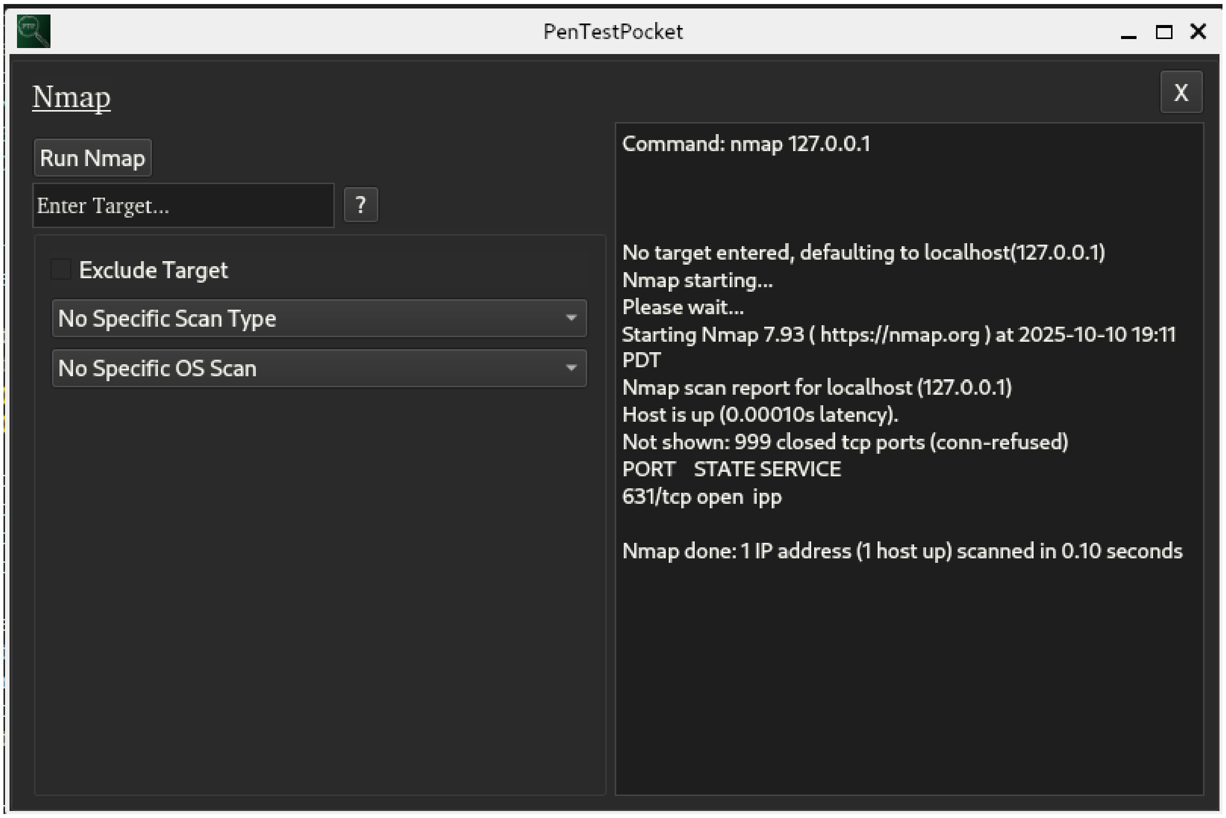
1a.

Base Case



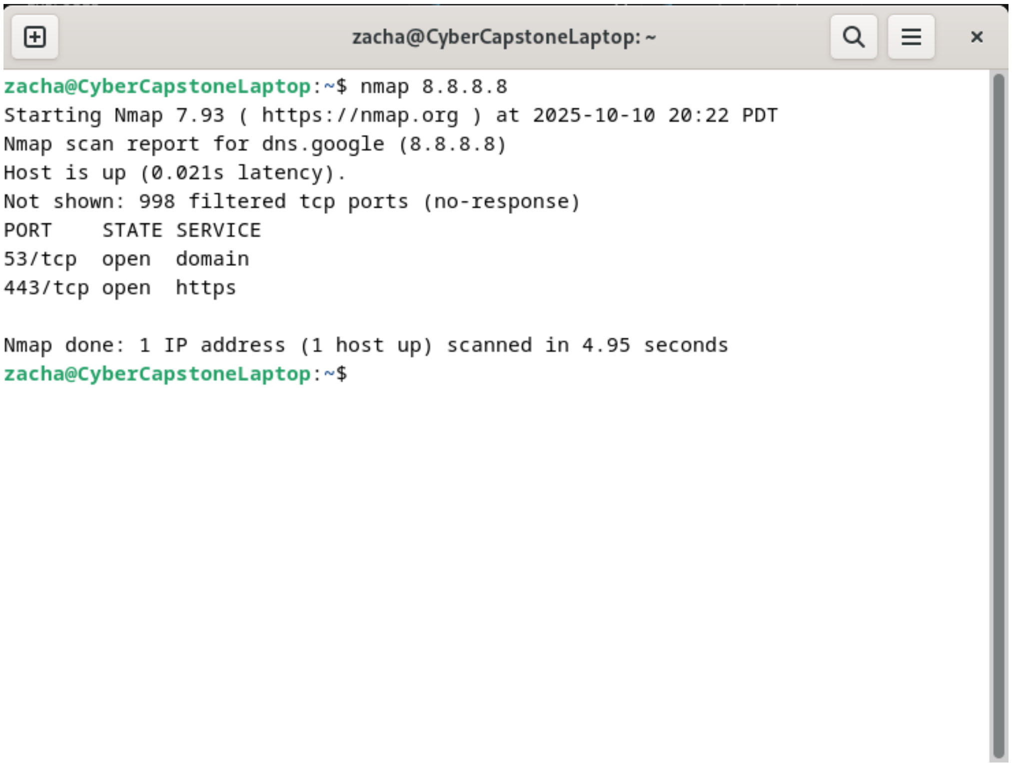
```
zacha@CyberCapstoneLaptop: ~/Capstone/Adelson_Senior_Capstone/Pen...
zacha@CyberCapstoneLaptop:~/Capstone/Adelson_Senior_Capstone/PenTestPocket$ nmap
 127.0.0.1
Starting Nmap 7.93 ( https://nmap.org ) at 2025-10-10 19:13 PDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000095s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT    STATE SERVICE
631/tcp open  ipp

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
zacha@CyberCapstoneLaptop:~/Capstone/Adelson_Senior_Capstone/PenTestPocket$
```

Test Case

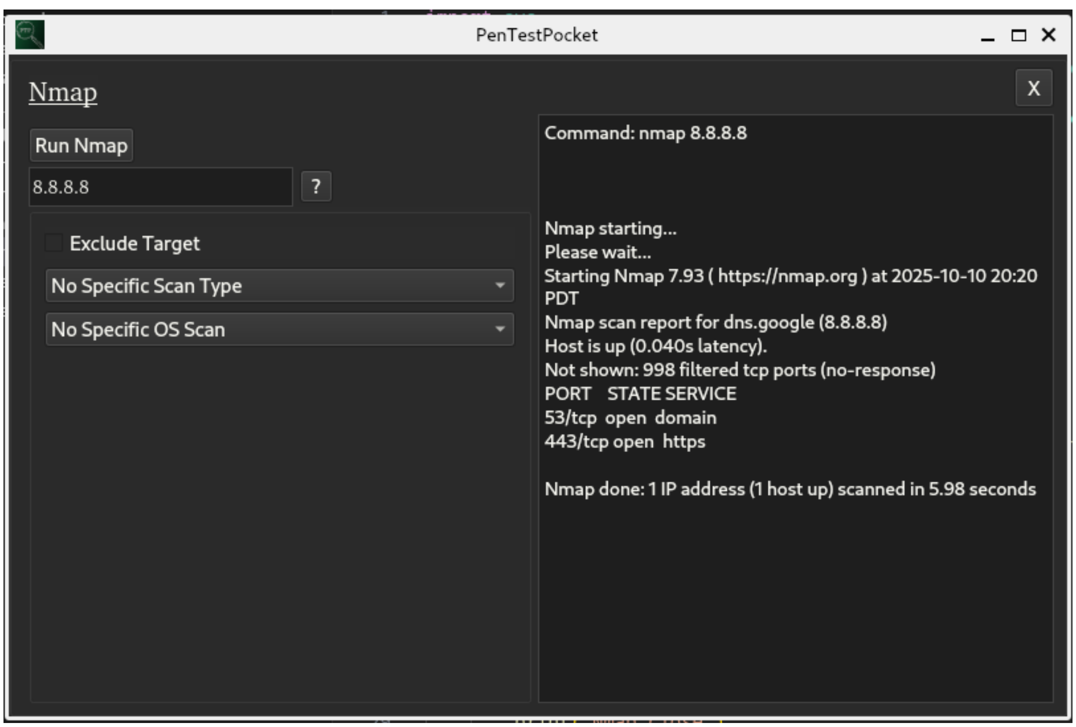

PenTestPocket
Nmap
X
Run Nmap
Enter Target...
?
Exclude Target
No Specific Scan Type
No Specific OS Scan
Command: nmap 127.0.0.1
No target entered, defaulting to localhost(127.0.0.1)
Nmap starting...
Please wait...
Starting Nmap 7.93 (https://nmap.org) at 2025-10-10 19:11 PDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00010s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT STATE SERVICE
631/tcp open ipp
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

1b.

Base Case

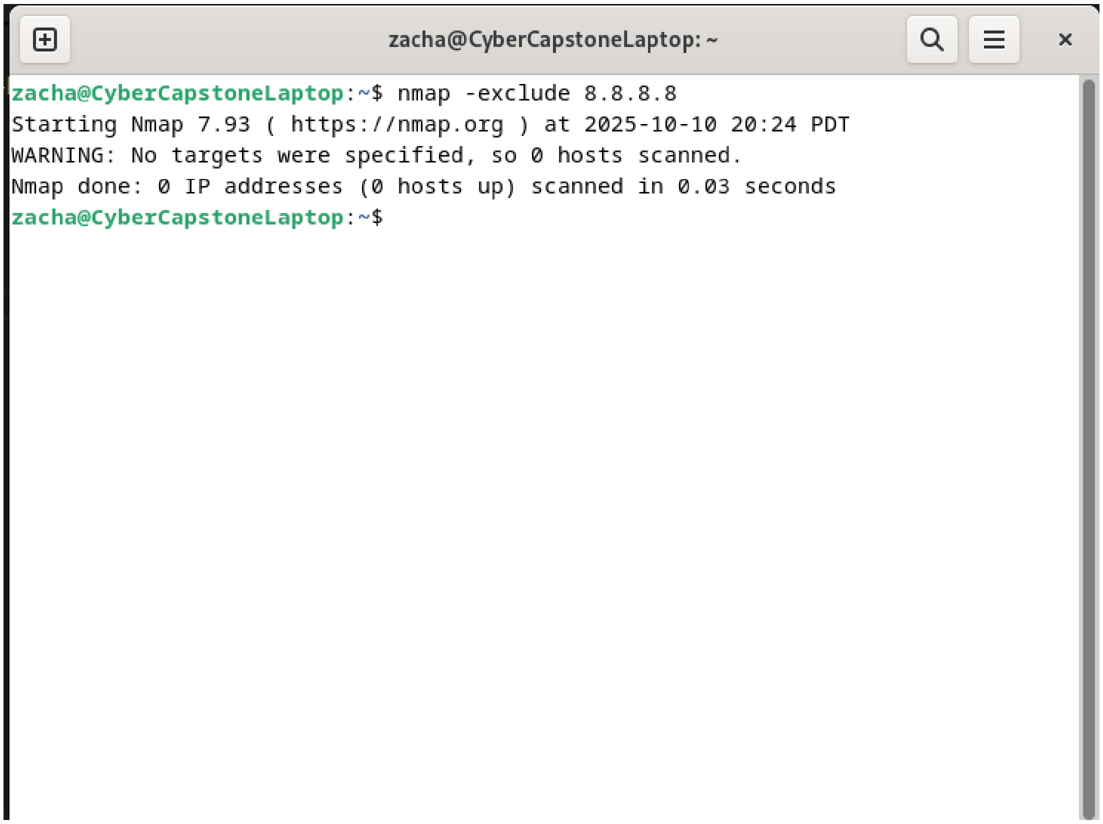


Test Case

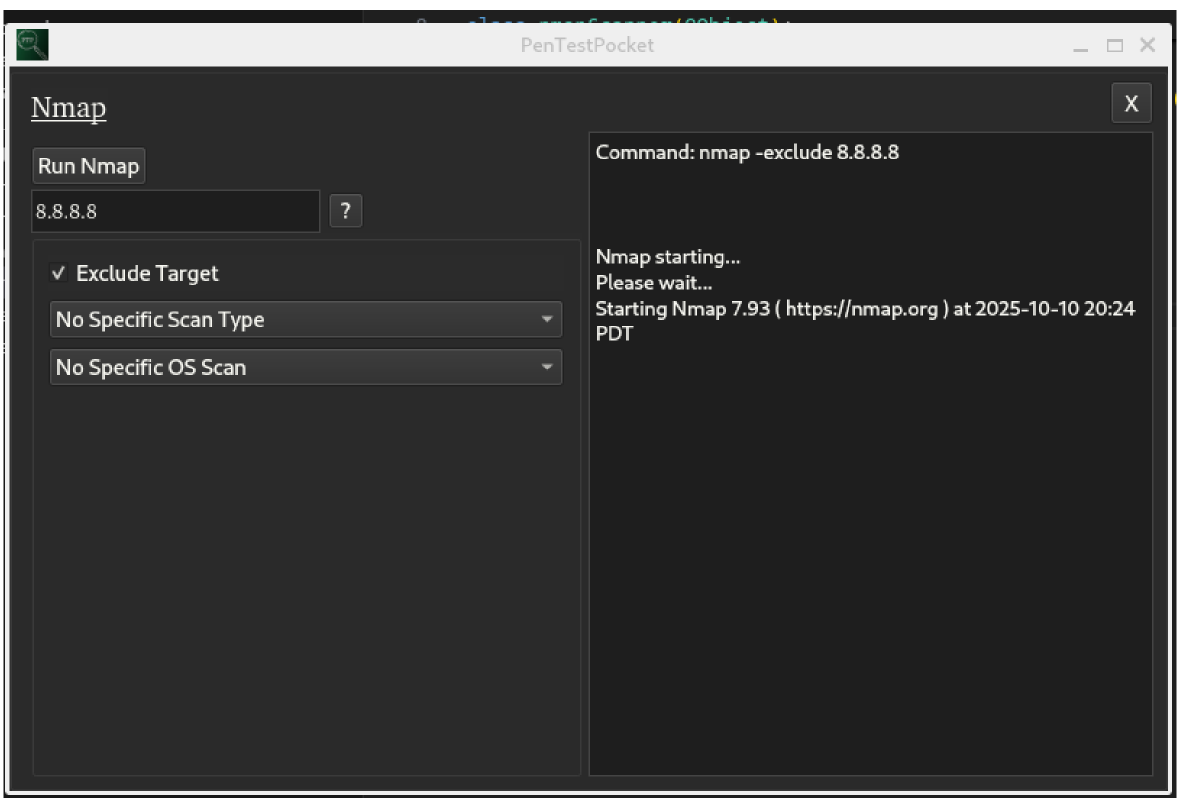


1c.

Base Case



Test Case



Projects and Tools used

Tool/library/framework	Designed Usage(s)
PyQt5	Designing GUI and allowing for simple, cross platform application creation and development
	Intercommunication between our modules

Languages Used in Project			
Python3	BASH shell	PowerShell	LaTeX
JSON [Not presently Used]			

V. Description of Final Prototype

[OUT OF SCOPE FOR SPRINT 2]

In this section provide a **detailed** description of your final prototype.

(If applicable) Include a brief user manual for your final software where you provide step by step instructions for using your system. Explain the major use case scenarios. You may include screenshots of the user interface.

This is the first section that is somewhat really new for this document
Describe your final prototype implementation. Please format this section according to what you think is the best way to describe your prototype. The following is just a suggestion.

I recommend to include plenty of images and pictures of the following where appropriate:
- any diagrams/figures that visualize various features of your prototype;
- the screenshots of your user interfaces;
- the screenshots of your test programs;
- pictures of your team testing and debugging the devices, programs, etc.

A well-thought and clear diagram is better than long and descriptive text.

If your document starts to be very long due to screenshots and diagrams, please put at least some of them into an appendix to this document.

VI. Social Responsibility and Broader Impacts

[OUT OF SCOPE FOR SPRINT 2]

Social responsibility: identify informed judgements that you made for this project based on legal and ethical principles. Discuss if those judgements are in line with your responsibilities as a computing / software / cyber engineer. If not, what prompted you to make judgements that are contradictory to your professional responsibilities?

Broader impacts: highlight how your work aligns with the WSU EECS's goals of benefiting society and broadening participation in STEM. Describe not only how your project addressed a real-world problem (briefly state the problem) but also emphasized accessibility, inclusivity, and/or educational outreach. If you engaged with local communities through user testing and feedback sessions, mention that. If you collaborated with multiple teams from various departments or groups, mention that. If you documented your project and released it freely to support open-source missions, mention that. If your project outcomes contribute to technological innovation and provide a foundation for future student-led research, community-based applications, mention that.

VII. Product Delivery Status

[OUT OF SCOPE FOR SPRINT 2]

When was/will the project be delivered to your client? Was it demonstrated and who did you hand it off to?

Also include project location information here. This is for both the instructor and your client as future reference. This should include:

1) Source repositories
2) Equipment storage location - where did you leave any client or course materials
3) Any other materials needed to rebuild your project from the ground up

Ensure that either here or in your detailed description you include any instructions or where the instructions are to install and setup your project so future users know where to start.

VIII. Conclusions and Future Work

VIII.1. Limitations and Recommendations

[OUT OF SCOPE FOR SPRINT 2]

Include a brief description of the limitations of your current prototype that you are aware of, and discuss possible solution approaches to overcome those limitations.

Please keep the discussion brief. 1 page text should be sufficient for this section.

Note that there might be overlap between the Limitations and Recommendations section and the Future Work section. “Future Work” should discuss the possible ways to extend the project, where as “Limitations and Recommendations” should be more detailed and should discuss the problems and missing features in the final prototype and should describe the possible solution approaches.

VIII.2. Future Work

Currently Irrelevant as project is still in development

IX. Acknowledgements

I would like to acknowledge and thank my professor and mentor Ananth Jillepalli for supporting my project and collaborating with me to get this project off the ground running. PTP was independently designed and thought up by me, and Ananth Jillepalli worked tirelessly to ensure I had approval from the department and that my project would make sense within the scope of a capstone project. I would also like to thank my professional sponsor Dayton Dekam from CrowdStrike for believing in my idea enough to function as the sponsor when the university was unable to gather sponsors for the senior capstone project course. Lastly, I would like to thank the departments of EECS (Electrical Engineering and Computer Science) and VCEA (Voiland College of Engineering and Architecture) for approving my project and allowing me to bring my vision into reality with this semester long project.

X. Glossary

[OUT OF SCOPE FOR SPRINT 2]

Define technical terms, acronyms, and anything else the reader might need to look up as they go through your document

XI. References

[OUT OF SCOPE FOR SPRINT 2]

Cite all your references here. For the papers you cite give the authors, the title of the article, the journal name, journal volume number, date of publication and inclusive page numbers. Giving only the URL for the journal is not appropriate.

Use either Chicago, IEEE, or ACM style citations

-- Note: you can find many articles on scholar.google.com, which includes a link for each article's citation in various formats.


[HTML] Smart eldercare in Singapore: Negotiating agency and apathy at the margins
L Kong, O Woods - Journal of aging studies, 2018 - Elsevier
Around the world, smart technologies are being embraced as a cost-efficient means of enabling the elderly to be cared for in new, more non-proximate ways. They can facilitate ageing-in-place, and have the potential to relieve pressure on the providers of care. Yet, the fact is that the interface of technology and society is a negotiated one. These negotiations are most acutely felt when technology is used to supplement the hitherto human-centred process of caregiving, especially amongst "marginalised" societal cohorts, like the elderly …
Cited by 1 Related articles All 6 versions

XII. Appendix A - Example Testing Strategy Reporting

When testing the goal is to ensure that buttons and code are executing how we expect, and that tools are returning information as the original tool does.

When testing tools, we will run the command in the terminal, screenshot the results, and run the command in PTP screenshot those results, and compare, and if they are identical, we know our test is successful

When testing code, we describe exactly what the code is intended to do, and then run those steps, executing the code, and document the output with screenshots

XIII. Appendix B - Project Management

Every week since this is a solo project is just independent work on the Capstone assignment. On Thursdays before the weekly mentor meeting, I will add new issues for the coming week and upload my time tracking excel sheet, which is used to tabulate all the time I have spent on the Capstone project each day, by categorization.

For project planning, I use GitHub issues and the GitHub roadmap grouped into 4 distinct groups; Logistics, Code, Meetings, and Reporting.

Logistics is for everything project management wise such as blocking time, planning next steps, etc.

Code is for the time spent working on the actual application side, including all time spent developing and testing PTP.

Meetings are reserved for the time spent during mentor meetings, which on a standard week is just 30 minutes on Thursdays.

Reporting is for any write-ups and documentation, including sprint and project reports; This is also meant for any instructions or documentation needed to view and use PTP.